

Civil Engineering

Calculus and Differential Methods

Trainer Answer Guide - Keep Private

This answer guide is separated from learner materials. Do not share publicly unless releasing solutions.

Suggested Marking Guidance

Award marks for correct formula selection, substitution, calculation accuracy, units, and technical interpretation.

$$\frac{d}{dx}(ax^n) = anx^{n-1}$$

$$\int ax^n dx = \frac{ax^{n+1}}{n+1} + C$$

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

1. Expected approach: identify data, choose formula, substitute correctly, compute accurately, and interpret in civil engineering context.
2. Expected approach: identify data, choose formula, substitute correctly, compute accurately, and interpret in civil engineering context.
3. Expected approach: identify data, choose formula, substitute correctly, compute accurately, and interpret in civil engineering context.